Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/28/2005
PI Name:	Aldredge, Ralph Ph.D.		
Project Title:	Flame Propagation in Low-Intensity Turbulence under Microgravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	aldredge@ucdavis.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	530-752-5016
Organization Name:	University of California, Davis		
PI Address 1:	Mechanical & Aeronautical Engineering		
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PI Web Page:			
City:	Davis	**State:**	CA
Zip Code:	95616-5294	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	12/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Miller, Fletcher J	**Contact Phone:**	619-594-5791
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NCC3-845		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Microgravity and Ground-based experimental studies of premixed-flame propagation in low-intensity turbulent flow under microgravity conditions are to be undertaken The focus of the research is on the hydrodynamic coupling between flame and flow-field dynamics at low turbulence intensities, which may lead to flame-generated turbulence, flame acceleration and enhancement of burning rates. As a result of hydrodynamic coupling the transition from initially planar or wrinkled laminar flames to fast turbulent flames or detonations is possible, even when diffusive-thermal effects associated with non-unity reactant Lewis numbers are not destabilizing. Therefore, the proposed work is relevant to microgravity fire safety.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	During the past year, we have completed construction of a new non-rotating burner to be used for ground-based and microgravity studies of the Darrieus-Landau hydrodynamic instability on annular-flame propagation. We have also developed and tested new optical tools to enable measurements of turbulence properties in a second, rotating burner to be under construction soon. Preliminary ground-based studies will be carried out at UC Davis over the next few months, after-which microgravity experiments at the NASA Glenn Research Center will be conducted.
Bibliography Type:	Description: (Last Updated: 04/13/2017)
Articles in Peer-reviewed Journals	Aldredge, R. C. "Saffman-Taylor influence on flame propagation in thermoacoustically excited flow" Combustion Science & Technology 177:53-73 , Nov-2004
Articles in Peer-reviewed Journals	Aldredge RC. "Methane-air Markstein numbers from measurements of thermoacoustic instability" Combustion Science and Technology. 2005;177(5-6): 1023-47. http://dx.doi.org/10.1080/00102200590926978 ; Originally reported in Dec 2005 as "accepted, December 2005" , Dec-2005